VeriScan: Deepfake Video Detection System

Introduction

Deepfakes are AI-generated manipulated videos that can spread misinformation, fraud, and fake evidence. VeriScan is an AI-powered solution that uses deep learning (CNN + LSTM) to detect such fake videos and safeguard trust in digital media.

Problem Statement

Traditional human inspection cannot reliably detect deepfakes. Manual methods are slow and error-prone, and existing detection tools lack accessibility and accuracy.

Objectives

- Detect fake vs real videos accurately
- Use CNN (Xception) for frame-level analysis
- Use LSTM for sequence-level inconsistencies
- Provide confidence score for predictions

Methodology

1. Data Collection: Deepfake datasets like FaceForensics++, Celeb-DF.
2. Preprocessing: Extract frames, crop faces, resize, normalize.
3. CNN (Xception): Extract frame features from each video frame.
4. LSTM: Feed CNN features as sequences to learn temporal inconsistencies.
5. Classification: Output Real/Fake with probability.

Architecture Workflow

Video → Frame Extraction → Face Detection → CNN (Xception for features) → LSTM (sequence analysis) → Output (Real/Fake + Confidence Score)

CNN and LSTM Integration

The CNN (Xception) is used as a feature extractor. Each frame from the video is resized to 299x299 pixels and passed through Xception to generate a 2048-dimensional feature vector. These feature vectors are then stacked as a sequence (e.g., 16 frames → 16x2048 matrix). This sequence is passed into the LSTM model, which analyzes temporal patterns like blinking, lip movement, and head motions. Finally, the LSTM outputs a probability score (0 = Real, 1 = Fake).

This combination allows VeriScan to detect both spatial (image-level) and temporal (sequence-level) artifacts.

Python Libraries and Their Role

- TensorFlow/Keras: Build CNN (Xception) and LSTM models, training and evaluation.
- OpenCV: Extract video frames, resize, preprocess images.
- MTCNN/Dlib: Detect and crop face regions in frames.
- NumPy/Pandas: Handle arrays and structured data for training.
- Matplotlib/Seaborn: Visualize training results, confusion matrix, ROC curves.
- fpdf/ReportLab: Export results and project reports in PDF format.

Model Evaluation

Performance Metrics:
- Accuracy
- Precision, Recall, F1-Score
- Confusion Matrix
- ROC-AUC curve

Applications

- Journalism: Verify authenticity of news videos
- Cybersecurity: Prevent fraud using fake videos
- Forensics: Verify legal evidence
- Social Media: Flag deepfake content

Challenges

- Deepfakes keep evolving
- Dataset bias may reduce accuracy
- Real-time detection is computationally expensive

Future Scope

- Audio deepfake detection
- Browser plugin for real-time video checks
- Mobile application for public use
- Multi-modal detection combining video + audio

Conclusion

VeriScan combines CNN (Xception) and LSTM to detect deepfakes effectively, ensuring authenticity in digital media. By leveraging Python libraries like TensorFlow, OpenCV, and MTCNN, the system integrates image processing and deep learning seamlessly. With further improvements, it can become a mainstream tool for cybersecurity, journalism, and legal systems.